

Question: 1

Prepare a 10-slide presentation that explains your IT infrastructure project, including the technical setup, business problem it solves, and key features. Use real-world examples from apps like Zomato, Flipkart, or Paytm to illustrate your points?

Answer: • Slide 1: Title Slide

- Title: Enterprise IT Infrastructure Design.
- Subtitle: High-Availability Network for Co-Working Spaces.
- Slide 2: Business Problem
 - Issues: High network lag, zero line redundancy, and security risks.
 - App Example: Like a crash on Zomato, local network downtime freezes business operations.
- Slide 3: Technical Setup Overview
 - Core Tier: Multi-tier model (Edge Routers → Core Switches → Access Switches).
 - Segmentation: Isolated Virtual Local Area Networks (VLANs).
- Slide 4: IP Subnetting & Security
 - VLANs: VLAN 10 (Marketing), VLAN 20 (Tech), and VLAN 30 (Finance).
 - App Example: Modeled after Paytm's secure payment layer to sandbox sensitive accounting data.
- Slide 5: Gateway Redundancy & Uptime
 - Protocol: Dual-WAN gateways running Virtual Router Redundancy Protocol (VRRP).
 - App Example: Mirrors Zomato's active backup system to guarantee 100% internet uptime.
- Slide 6: Bandwidth Management
 - Control: Strict Quality of Service (QoS) rules on Layer 3 core switches.

- App Example: Mimics Flipkart's traffic scaling during big sales to prevent system crashes.
- Slide 7: Hardware Bill of Materials (BOM)
 - Routers: 2x TP-Link ER605 Gateways — ₹15,600
 - Switches: 2x Cisco CBS250 Layer 3 Switches — ₹49,000
 - Access Points: 3x Aruba Instant On AP11 Units — ₹56,997
 - Cables: 1x D-Link Cat6 Cable Drum — ₹9,200
 - Total Budget: ₹1,30,797
- Slide 8: Wireless Hardening & Security
 - Staff WiFi: Secure authentication using advanced WPA3-Enterprise protocols.
 - Guest WiFi: Isolated Guest VLAN managed by an OTP captive portal.
- Slide 9: Post-Review Physical Revisions
 - Enclosures: Added lockable anti-tamper security boxes for public Access Points.
 - User Experience: Extended guest portal login sessions to a full 8-hour workday lease.
- Slide 10: Conclusion & Scalability
 - Status: Fully delivers high availability, security boundaries, and cost tracking.
 - Future Scaling: Transitioning from /24 to /23 block sizes to support up to 510 hosts.

Question: 2

Write a one-page technical summary document for your project, clearly describing the main components, technologies used, and how they work together to deliver the solution.

Hint: Focus on clarity and include diagrams if possible.

Answer: • Architecture Model: Designed a resilient multi-tier hierarchical infrastructure layout.

- Network Isolation: Configured independent Virtual Local Area Networks (VLANs).
- Edge Routing: Implemented 2x TP-Link ER605 Wired Multi-WAN Routers.
- Switching Fabric: Selected 2x Cisco Business CBS250 Layer 3 Switches.
- Wireless Access: Distributed 3x Aruba Instant On AP11 Access Points.
- Physical Connections: Used full gigabit D-Link Cat6 UTP Copper Cables.
- Gateway Redundancy: Deployed VRRP protocol across dual ISP connections.
- Loop Prevention: Activated RSTP configuration across stacked switch trunks.
- Security Boundaries: Enforced Layer 3 Access Control Lists (ACLs).
- Traffic Optimization: Applied QoS rules to prioritize corporate applications.
- User Access: Secured internal endpoints via WPA3-Enterprise protocols.
- Guest Control: Isolated public clients using an OTP Captive Portal.
- Physical Hardening: Enclosed public access points in lockable security boxes.
- Future Scalability: Formulated expansion paths to shift into larger /23 block sizes.

Question: 3

List three questions you think the evaluation panel might ask about your project's technical correctness or business alignment, and write a brief answer for each.

Hint: Consider questions about scalability, security, or how your project supports a business goal.?

Answer: • Question 1 (Focus: Scalability)

- Panel Question: How does your current network architecture accommodate a sudden increase in tenant numbers or devices within the co-working space?
- Brief Answer: The infrastructure uses modular VLAN segmentation paired with a Class C private IP addressing scheme. While each department is currently capped at 254 endpoints using a /24 subnet mask, the system is designed to scale up to 510 hosts per domain by seamlessly shifting to a broader /23 block configuration without requiring new hardware.
- Question 2 (Focus: Security)
- Panel Question: How does your design ensure that sensitive financial data cannot be accessed by external clients or other tenants sharing the space?
- Brief Answer: Complete logical boundary isolation is enforced using strict Layer 3 Access Control Lists (ACLs) on the core switches. This completely blocks traffic from the general tenant or guest networks from routing into the dedicated Finance VLAN (VLAN 30) domain, mimicking enterprise-level financial data security.
- Question 3 (Focus: Business Alignment)
- Panel Question: How does this network design protect the co-working business from financial losses caused by internet or hardware downtime?
- Brief Answer: The design integrates a Dual-WAN gateway configuration using VRRP and stacked Layer 3 core switches running RSTP. This setup eliminates single points of failure by automatically failing over to a backup ISP line or standby switch within milliseconds, guaranteeing 100% network uptime to preserve tenant satisfaction and business revenue.

Question: 4

Review your project documentation and checklist to ensure every required feature is implemented and documented. Mark any incomplete sections and write a short plan for how you would finish them if given more time?

Answer: Project Documentation Review & Checklist

- Status of Required Features:
- Network Topology Diagram: Completed. (Hierarchical multi-tier layout is fully mapped).
- IP Addressing Sheet: Completed. (Class C subnets mapped for all departments).
- Redundancy & Failover Plan: Completed. (VRRP and RSTP protocols are fully integrated).
- Bill of Materials (BOM): Completed. (Hardware quantities and Indian market prices listed).
- Network Security Policy: Completed. (WPA3, ACL boundaries, and guest portal set up).

Incomplete Sections & Future Completion Plan

If given more time, the following technical sections would be fully developed to upgrade the project to an enterprise-production level:

- Incomplete Section 1: Detailed Router/Switch Configuration Scripts
- The Gap: The current documentation describes the routing protocols (VRRP, RSTP, ACLs) conceptually but lacks the exact command-line interface (CLI) scripts.
- Completion Plan: I would write out the exact Cisco IOS and TP-Link Omada CLI commands for creating the VLAN interfaces, assigning port security rules, and binding the Access Control Lists to the Finance VLAN.
- Incomplete Section 2: Comprehensive Network Monitoring System (NMS) Plan
- The Gap: There is no specific setup details for tracking network health or bandwidth spikes in real-time.
- Completion Plan: I would integrate a monitoring framework using open-source tools like Zabbix or PRTG. This would involve setting up SNMP (Simple Network Management Protocol) alerts to notify administrators immediately if an access point drops or if bandwidth usage crosses 90%.
- Incomplete Section 3: Physical Rack Layout and Cable Management Map

- The Gap: The physical arrangement of devices inside the main IT server rack is not visually mapped.
- Completion Plan: I would use a tool like MS Visio to design a 2D Server Rack Elevation Diagram. This would map out the exact placement of the patch panels, dual UPS battery units, switches, and routers from top to bottom, including color-coded cable run lines.

Question: 5

After your presentation, write a reflection (5-7 sentences) describing the feedback you received from the panel and what specific improvements you would make to your project or presentation skills based on that feedback?

Answer:1. The feedback helped me understand the strengths and weaknesses of my presentation.

2. I learned that my project was clear and informative.

3. I need to improve my confidence while speaking.

4. I would make my presentation more organized and clear.

5. I would maintain better eye contact with the audience.

6. I will practice more to improve my presentation skills.